



UNIVERSITY OF
ILLINOIS
URBANA-CHAMPAIGN

SE 423: Introduction to Mechatronics

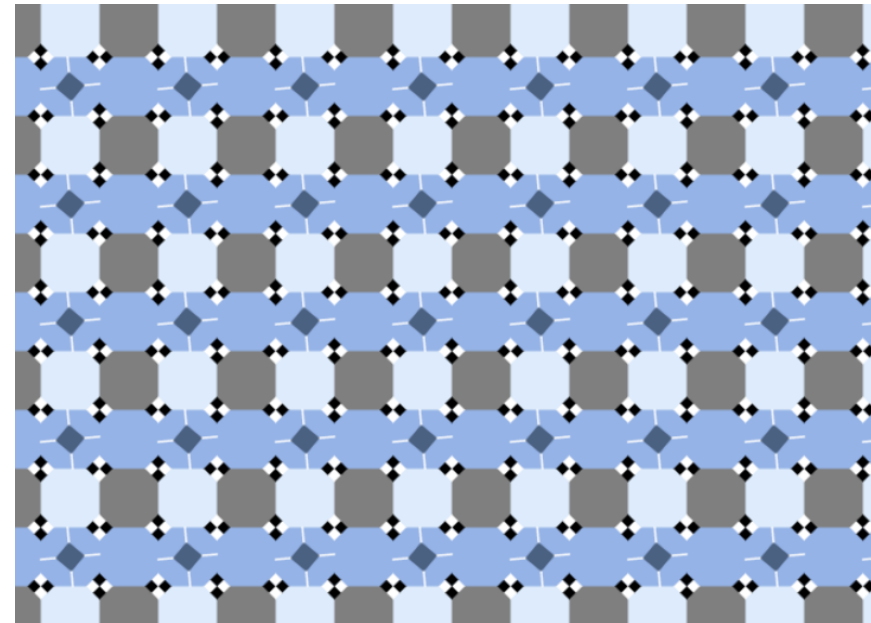
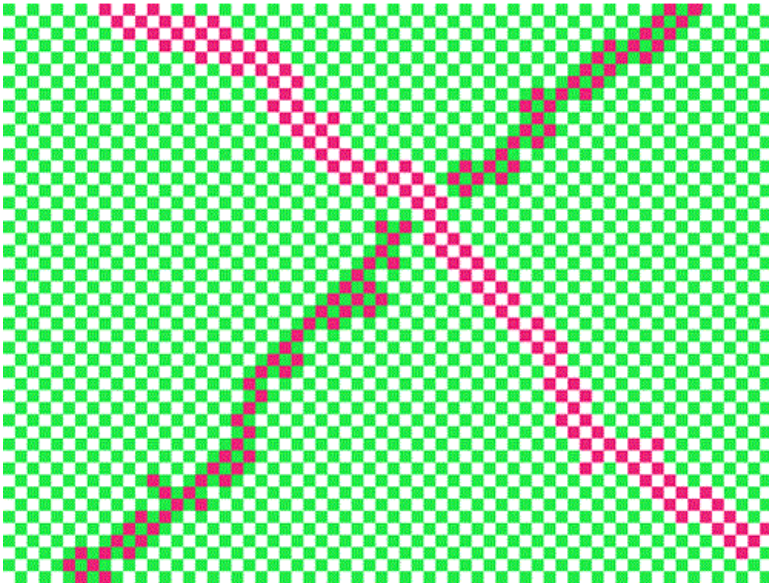
Lecture 17: Vision Processing, Cameras, Blob Detection

Marius Juston

DATE

Talk to the person next to you and answer these questions.

- What does RGB stand for?
- How many colors are there? Are the lines straight?



check out <https://michaelbach.de/ot/> for a bunch of optical illusions!

Office Hours:

- **Monday:** 4-6 PM, Neel
- **Tuesday:** 11-1 PM, Lakshmi
- **Tuesday:** 1-3 PM, Samuel
- **Tuesday:** 2-5 PM, Dan & Marius

Lab: Starting Lab 6 this week. This a 3-week lab!

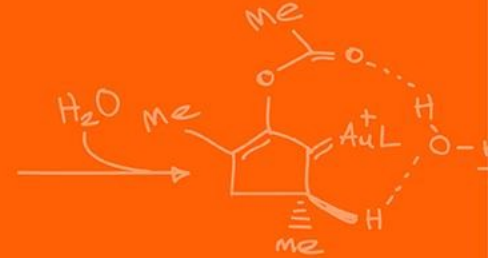
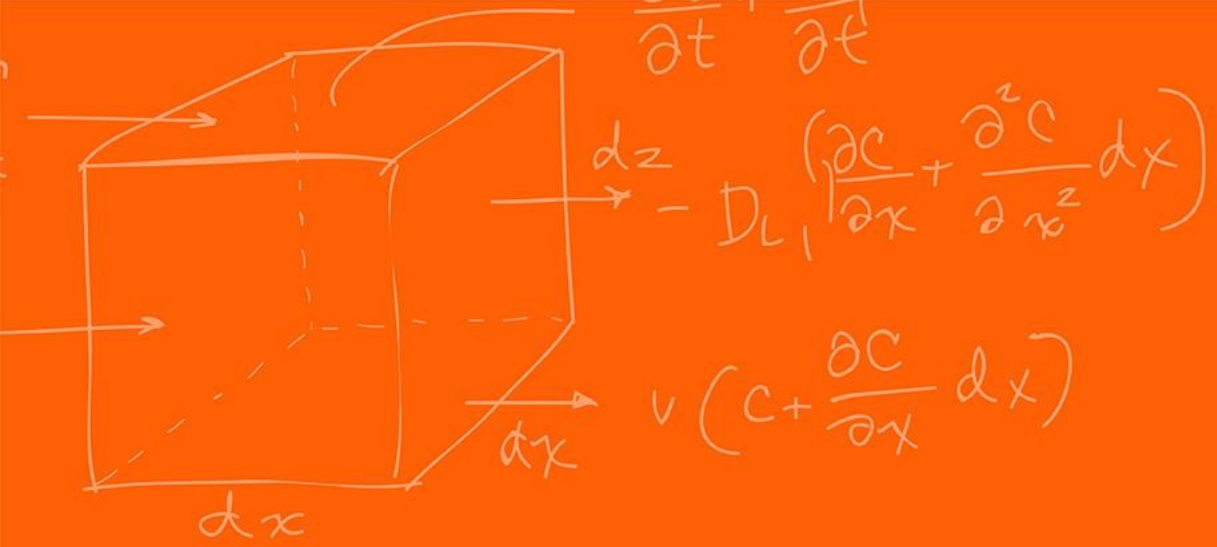
- Don't forget to have the Lab LabVIEW application implemented before lab.

Homeworks:

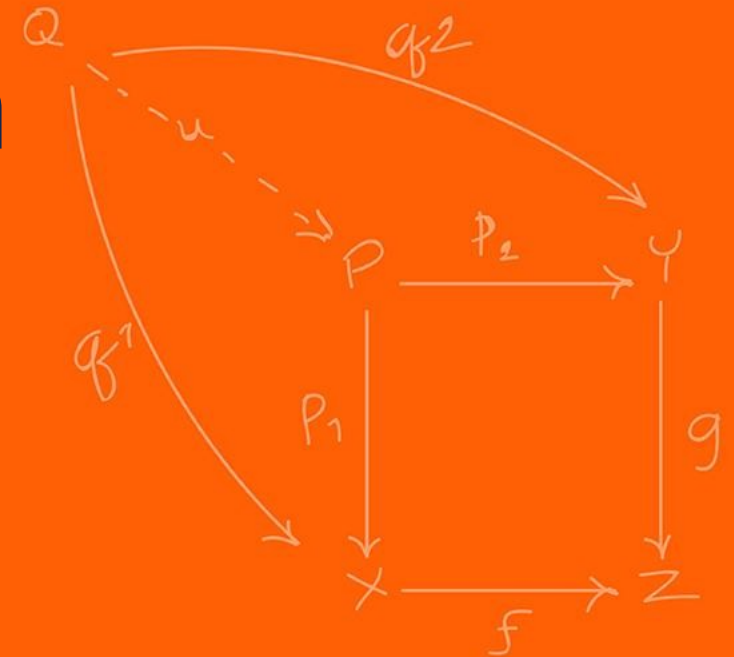
- Check-offs for [homework 4](#) Ex 2 and 3 are due **March 31, 5PM (The end of office hours)**
- Check-offs for [homework 4](#) Ex 4 are due **April 7, 5PM (The end of office hours)**
- Remainder of [homework 4](#) due on Gradescope on **April 8, 9AM**

Questions?

Homework, Lab, LabVIEW?

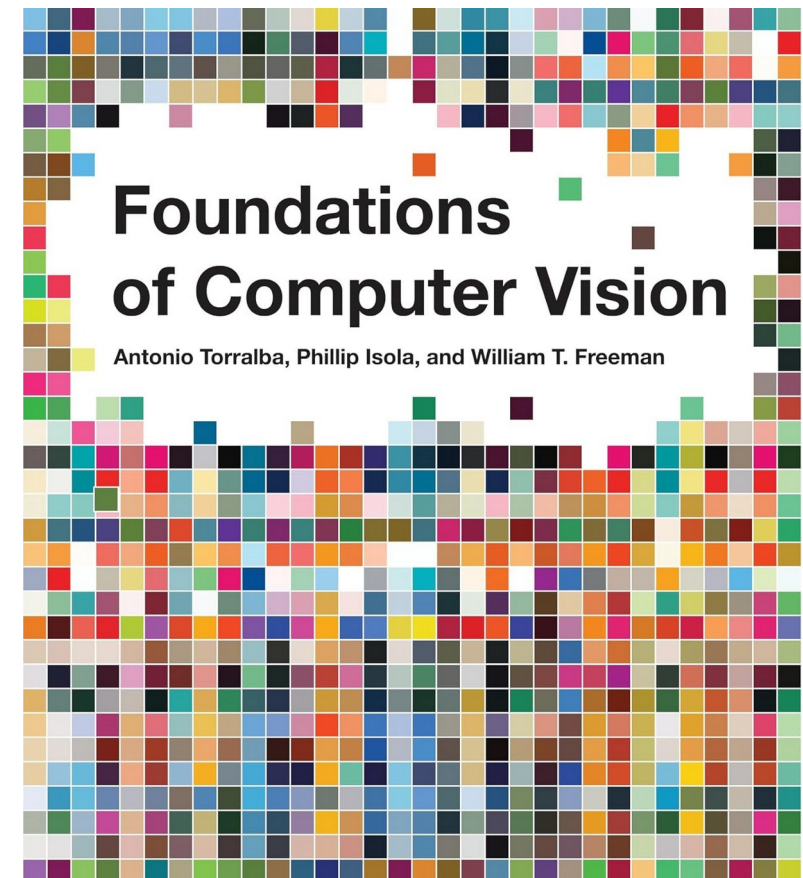


Computer Vision



I would **HIGHLY** recommend you looking at:

- <https://visionbook.mit.edu/>
- The webpage is a textbook with all the math, background and information for computer vision.
- We will only be doing a tiny portion of this in this lecture (there are whole grad classes for computer vision).
- It is containerized very nicely, has lots of pictures, diagrams. Each section takes 5 minutes to read.



Currently we have the LiDAR working and returning points; however, what kind of data are missing from just the LiDAR?

Color, complex shapes, human vision like data

Vision processing allows a robot to extract rich, high-density data from its environment.

Instead of an array of 228 distances (like our LiDAR), a camera gives us millions of data points per frame (RGB per pixel).

What is the primary drawback of having millions of data points per frame in a real-time mechatronic system?

Computational latency; processing takes time, more memory / bandwidth is required to handle processing of the data!

Is a standard camera an active or a passive sensor?

Passive.

It does not emit energy; it captures ambient light bouncing off objects.

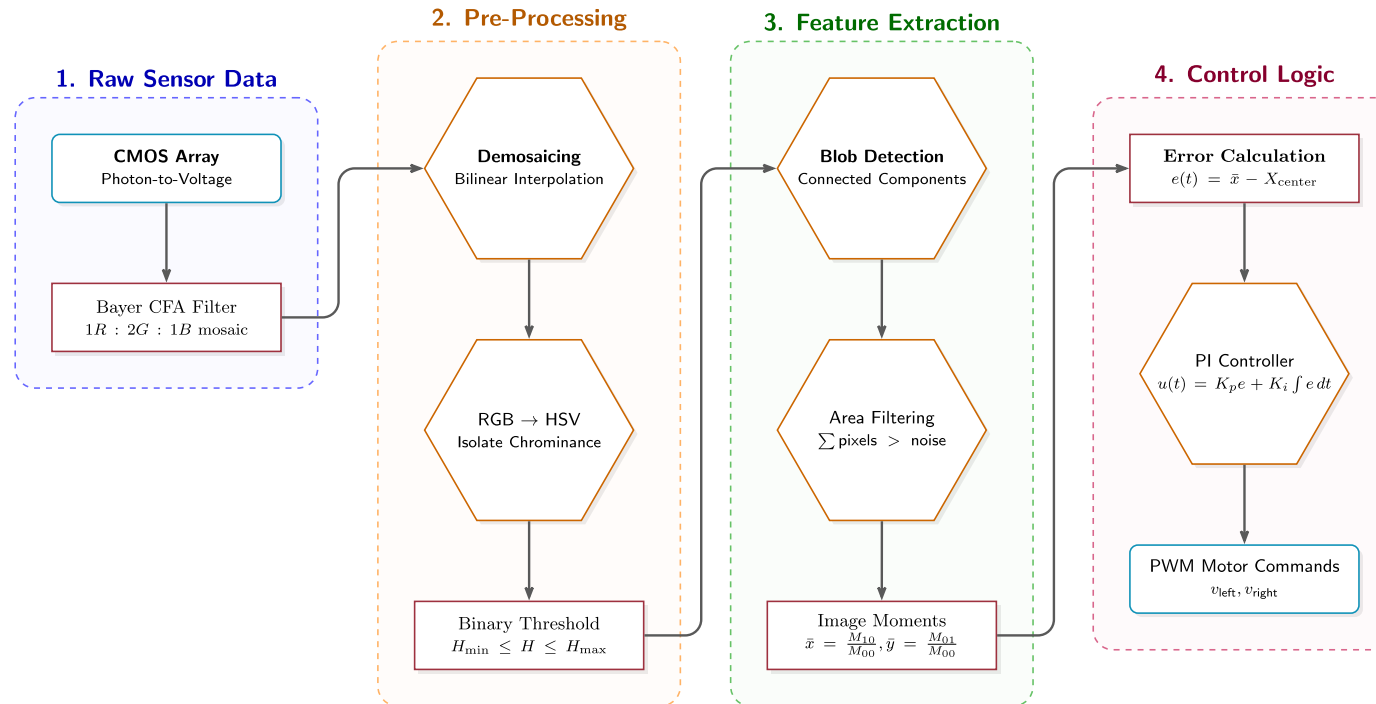
Because it is passive, it is highly vulnerable to environmental changes.

If the sun goes behind a cloud what happens?

The pixel color changes, shadows appears, etc.

How do we build robust code around this?

The main pipeline that we will follow in this class is the following:



Step 1: The CMOS sensor captures photons and converts them to digital arrays.

Step 2: The processor filters the image (e.g., finding specific colors).

Step 3: The algorithm calculates the coordinates of the target (Centroid).

Step 4: The Cascade Control loop turns the coordinates into wheel velocities.

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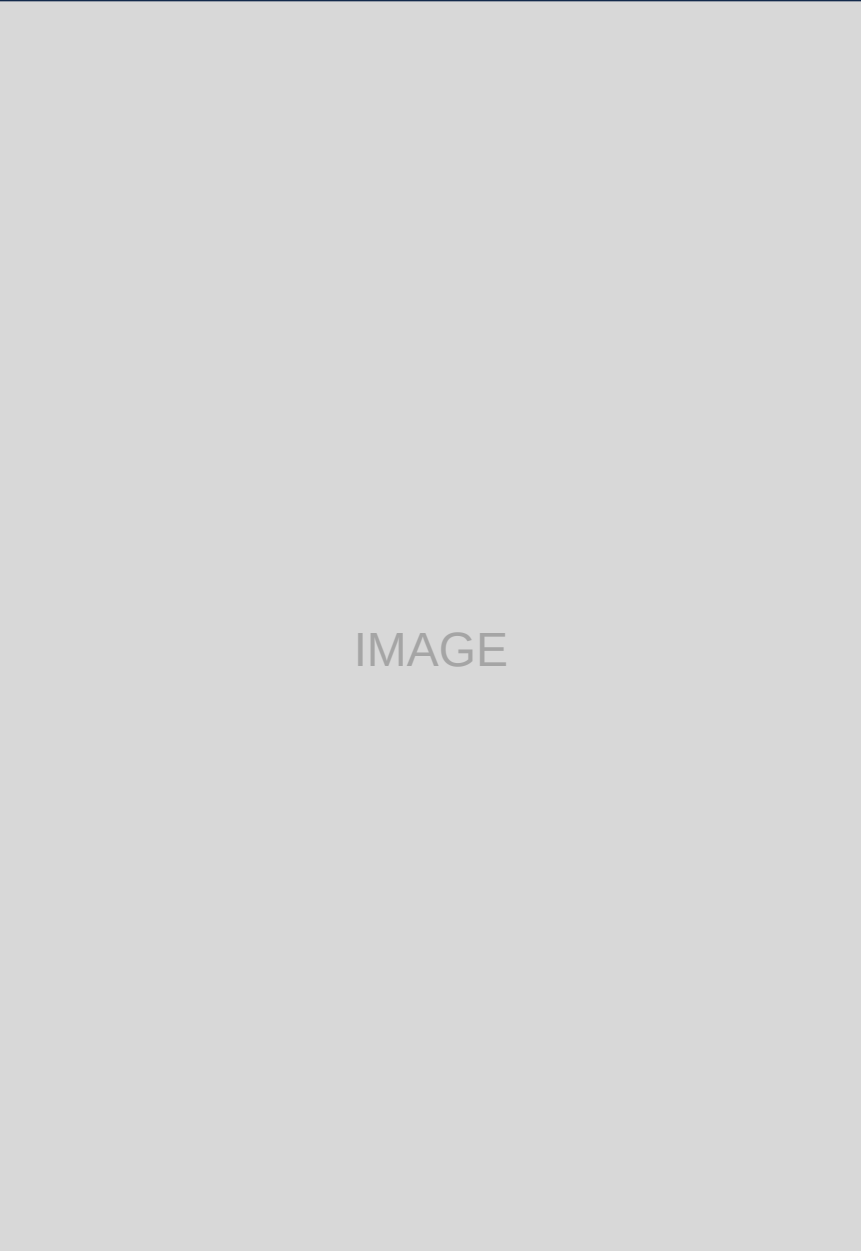


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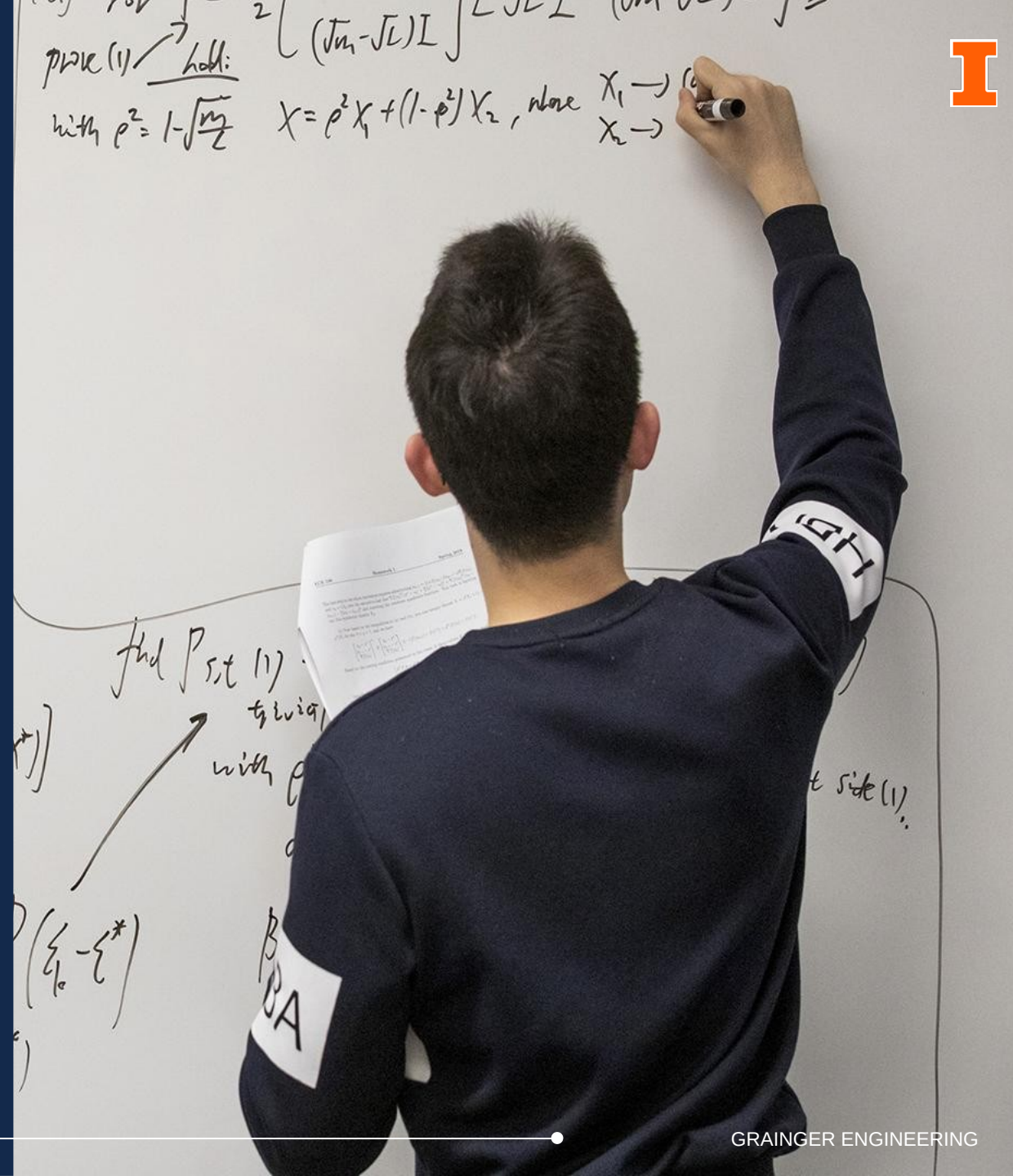
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IMAGE

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Summary

Thank You

Questions

Contact Information

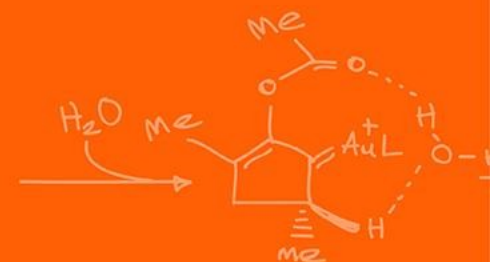
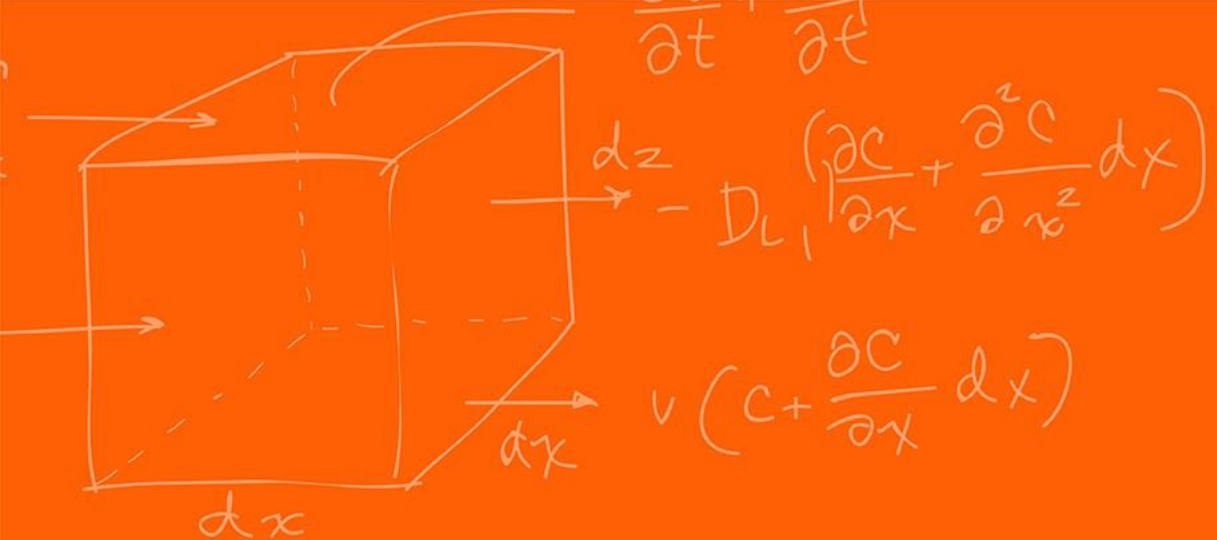
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Questions?



The Grainger College of Engineering

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